E. Long Permutation

time limit per test: 4 seconds
memory limit per test: 256 megabytes

A permutation is a sequence of integers from 1 to n of length n containing each number exactly once. For example, $[1]$, $[4, 3, 5, 1, 2]$, $[3, 2, 1]$ — are permutations, and $[1, 1]$, $[4, 3, 1]$, $[2, 3, 4]$ — no.

Permutation a is lexicographically smaller than permutation b (they have the same length n), if in the first index i in which they differ, $a[i] < b[i]$. For example, the permutation $[1, 3, 2, 4]$ is lexicographically smaller than the permutation $[1, 3, 4, 2]$, because the first two elements are equal, and the third element in the first permutation is smaller than in the second.

The next permutation for a permutation a of length n — is the lexicographically smallest permutation b of length n that lexicographically larger than a. For example:

- for permutation $[2, 1, 4, 3]$ the next permutation is $[2, 3, 1, 4]$;
- for permutation $[1, 2, 3]$ the next permutation is $[1, 3, 2]$;
- for permutation $[2, 1]$ next permutation does not exist.

You are given the number n — the length of the initial permutation. The initial permutation has the form $a = [1, 2, \ldots, n]$. In other words, $a[i] = i$ $(1 \le i \le n)$.

You need to process q queries of two types:

- $1\ l\ r$: query for the sum of all elements on the segment $[l, r]$. More formally, you need to find $a[l] + a[l + 1] + \ldots + a[r]$.
- $2\ x$: x times replace the current permutation with the next permutation. For example, if $x = 2$ and the current permutation has the form $[1, 3, 4, 2]$, then we should perform such a chain of replacements $[1, 3, 4, 2] \to [1, 4, 2, 3] \to [1, 4, 3, 2]$.

For each query of the 1-st type output the required sum.

Input

The first line contains two integers n $(2 \le n \le 2 \cdot 10^5)$ and q $(1 \le q \le 2 \cdot 10^5)$, where n — the length of the initial permutation, and q — the number of queries.

The next q lines contain a single query of the 1-st or 2-nd type. The 1-st type query consists of three integers 1, l and r $(1 \le l \le r \le n)$, the 2-nd type query consists of two integers 2 and x $(1 \le x \le 10^5)$.

It is guaranteed that all requests of the 2-nd type are possible to process.

Output

For each query of the 1-st type, output on a separate line one integer — the required sum.

Example

input Copy

```
4 4
1 2 4
2 3
1 1 2
1 3 4
```

output Copy

```
9
4
6
```

Note

Initially, the permutation has the form $[1, 2, 3, 4]$. Queries processing is as follows:

1. $2 + 3 + 4 = 9$;
2. $[1, 2, 3, 4] \to [1, 2, 4, 3] \to [1, 3, 2, 4] \to [1, 3, 4, 2]$;
3. $1 + 3 = 4$;
4. $4 + 2 = 6$

Codeforces Round 681 (Div. 2, based on VK Cup 2019-2020 - Final)

Finished

Practice

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Choose file: Choose File No file chosen

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Submission	Time	Verdict
379175753	Jun/16/2026 22:56	Accepted

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→ Contest materials

- Tutorial

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